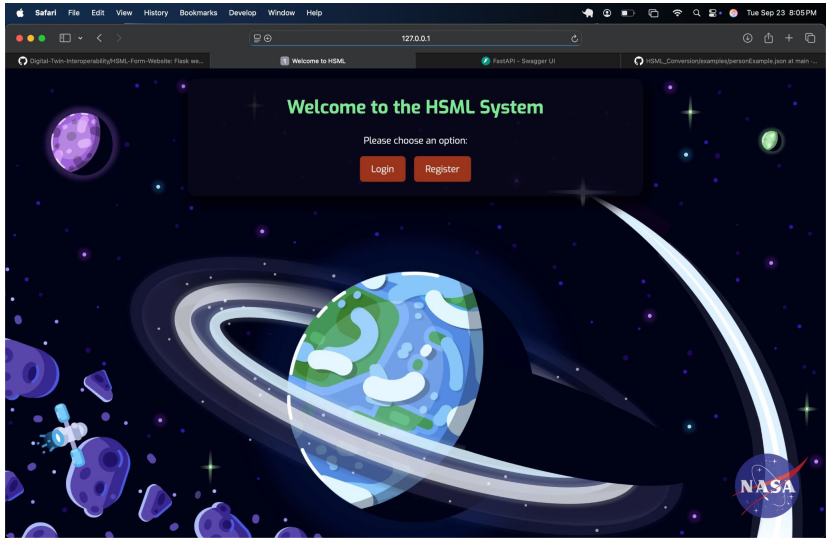
MoonWalker Weekly Meeting

9/17 - 9/24 Summary

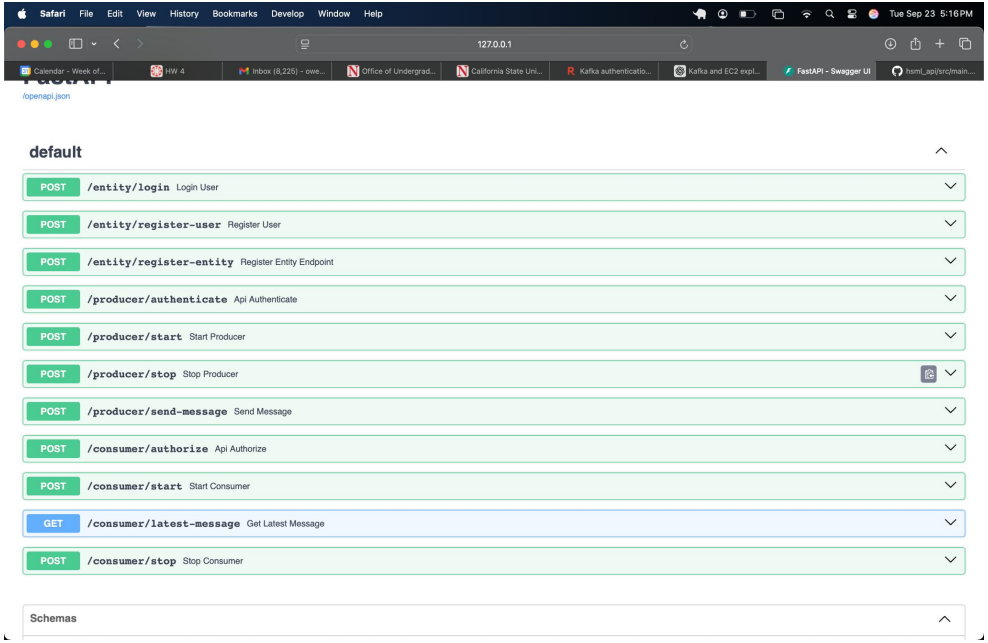
Elijah, Rowel, Martin

Progress Update

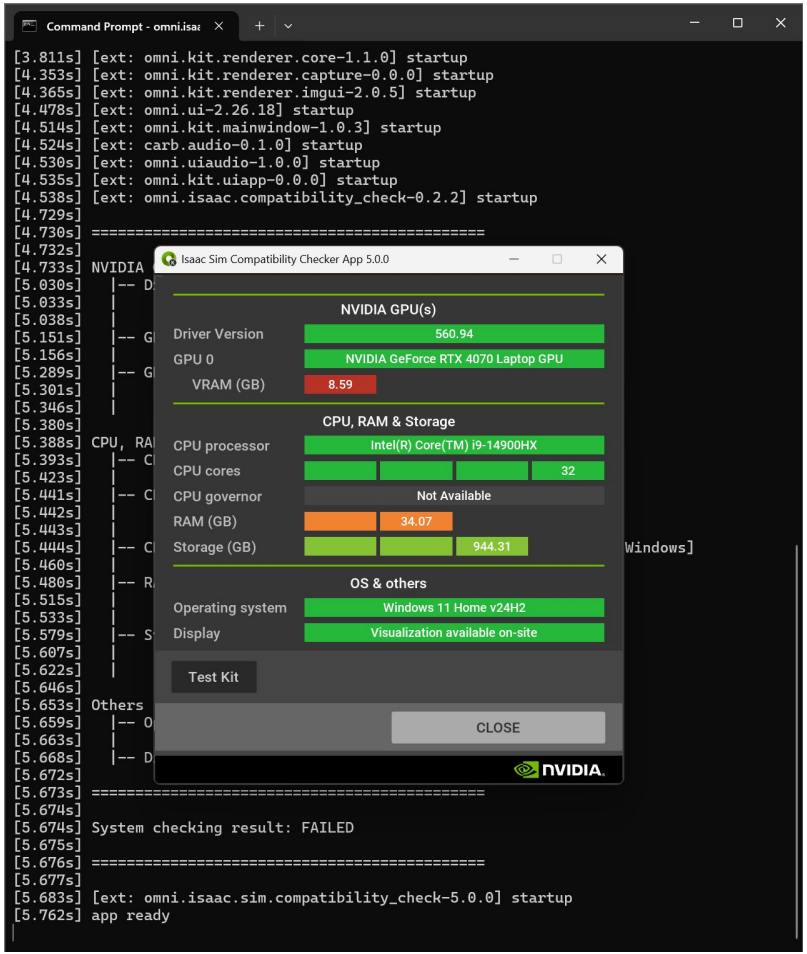
- Reviewed GitHub repositories
  - documentation
  - hsml_api
  - hsml_schema
  - HSML-Models
- Rowel:
  - HSML API locally setup
  - HSML Web-Form app locally setup
- Elijah:
  - Omniverse Isaac Sim locally setup
    - Compatibility check and stress tested
- Martin:
  - Researching how to implement Kafka w/ EC2 server
  - Learning/Familiarizing with Unity engine & C#
- Documenting our work
  - e.g. API setup, environment setup, other relevant instructions
  - Established guidelines/expectations for each other; accountability
- General Research
  - Kafka, EC2, Docker, Mysql db



Web form application (Login/Register)



FastAPI Swagger UI



Omniverse Compatibility Check

Startup Time

```
[45.857s] app ready
```

```
|-----------------------------------------------------|
|                    Summary Report                   |
|-----------------------------------------------------|
| workflow_name: benchmark_scene_loading              |
| env_url: /Isaac/Environments/Simple_Warehouse/full_warehouse.usd |
| num_cpus: 32                                        |
| gpu_device_name: NVIDIA GeForce RTX 4070 Laptop GPU |
|-----------------------------------------------------|
| Phase: loading                                      |
| System Memory RSS: 11.213 GB                        |
| System Memory VMS: 17.257 GB                        |
| System Memory USS: 10.713 GB                        |
| GPU Memory Tracked: 1.754 GB                        |
| Runtime: 18066.544 ms                               |
|-----------------------------------------------------|
| Phase: benchmark                                    |
| System Memory RSS: 11.282 GB                        |
| System Memory VMS: 17.325 GB                        |
| System Memory USS: 10.774 GB                        |
| GPU Memory Tracked: 1.754 GB                        |
| Mean FPS: 178.891 FPS                               |
| Real Time Factor: 1.035                             |
| Runtime: 9649.613 ms                                |
| Frametimes (ms):    mean |  stdev |   min |   max |
| App_Update          5.59 |   0.35 |  4.83 |  6.32 |
| Physics             1.33 |   0.13 |  1.10 |  1.61 |
|-----------------------------------------------------|
[433.370s] Simulation App Shutting Down
```

Full Warehouse Sample Scene Test

```
|-----------------------------------------------------|
|                    Summary Report                   |
|-----------------------------------------------------|
| workflow_name: benchmark_scene_loading              |
| env_url: /Isaac/Samples/ROS2/Scenario/carter_warehouse
| num_cpus: 32                                        |
| gpu_device_name: NVIDIA GeForce RTX 4070 Laptop GPU |
|-----------------------------------------------------|
| Phase: loading                                      |
| System Memory RSS: 4.727 GB                         |
| System Memory VMS: 8.758 GB                         |
| System Memory USS: 4.154 GB                         |
| GPU Memory Tracked: 1.754 GB                        |
| Runtime: 31427.048 ms                               |
|-----------------------------------------------------|
| Phase: benchmark                                    |
| System Memory RSS: 9.161 GB                         |
| System Memory VMS: 14.408 GB                        |
| System Memory USS: 8.526 GB                         |
| GPU Memory Tracked: 3.829 GB                        |
| Mean FPS: 10.086 FPS                                |
| Real Time Factor: 0.031                             |
| Runtime: 322133.187 ms                              |
| Frametimes (ms):    mean |  stdev |   min |   max |
| App_Update         99.15 |   2.05 | 94.65 | 103.01 |
| Physics             8.22 |   0.54 |  7.31 |  9.42 |
|-----------------------------------------------------|
[401.318s] Simulation App Shutting Down
```

Isaac ROC Sample Scene Test

```
|-----------------------------------------------------|
|                    Summary Report                   |
|-----------------------------------------------------|
| workflow_name: benchmark_o3dyn_robot                |
| num_robots: 10                                      |
| num_gpus: 1                                         |
| physics_backend: torch                              |
| num_cpus: 32                                        |
| gpu_device_name: NVIDIA GeForce RTX 4070 Laptop GPU  |
|-----------------------------------------------------|
| Phase: loading                                      |
| System Memory RSS: 9.574 GB                         |
| System Memory VMS: 13.821 GB                        |
| System Memory USS: 8.773 GB                         |
| GPU Memory Tracked: 1.768 GB                        |
| Runtime: 65845.36 ms                                |
|-----------------------------------------------------|
| Phase: benchmark                                    |
| System Memory RSS: 9.583 GB                         |
| System Memory VMS: 13.816 GB                        |
| System Memory USS: 8.777 GB                         |
| GPU Memory Tracked: 1.768 GB                        |
| Mean FPS: 17.106 FPS                                |
| Real Time Factor: 0.284                             |
| Runtime: 35252.284 ms                               |
| Frametimes (ms):    mean |  stdev |   min |   max  |
| App_Update         58.46 |   3.33 | 50.60 | 66.27  |
| Physics            33.62 |   2.44 | 27.21 | 38.62  |
|-----------------------------------------------------|
[139.850s] Simulation App Shutting Down
```

Physics Steps Per Second Test

```
|-----------------------------------------------------|
|                    Summary Report                   |
|-----------------------------------------------------|
| workflow_name: benchmark_sdg                        |
| num_frames: 600                                     |
| num_cameras: 2                                      |
| width: 1280                                         |
| height: 720                                         |
| asset_count: 100                                    |
| annotators: rgb, distance_to_image_plane            |
| num_gpus: 1                                         |
| num_cpus: 32                                        |
| gpu_device_name: NVIDIA GeForce RTX 4070 Laptop GPU  |
|-----------------------------------------------------|
| Phase: loading                                      |
| System Memory RSS: 4.526 GB                         |
| System Memory VMS: 9.455 GB                         |
| System Memory USS: 4.063 GB                         |
| GPU Memory Tracked: 2.32 GB                         |
| Runtime: 37614.251 ms                               |
|-----------------------------------------------------|
| Phase: benchmark                                    |
| System Memory RSS: 4.554 GB                         |
| System Memory VMS: 9.396 GB                         |
| System Memory USS: 4.084 GB                         |
| GPU Memory Tracked: 2.256 GB                        |
| Mean FPS: 0.758 FPS                                 |
| Real Time Factor: 0.013                             |
| Runtime: 793644.248 ms                              |
| Frametimes (ms):    mean |  stdev |   min |   max  |
| App_Update       1319.43 |  55.39 | 1181.86 | 1406.52 |
| Physics             2.19 |   0.23 |  1.65 |  2.63  |
|-----------------------------------------------------|
```

Synthetic Data Generation Test

Challenges

- Faced issues with Docker; had to set up environment without it
- Waiting for final decision on Kafka server placement
- Unable to connect to local MySQL database

```
%3|1758680601.440|FAIL|rdkafka#producer-2| [thrd:localhost:9092/bootstrap]: l
ocalhost:9092/bootstrap: Connect to ipv6#[::1]:9092 failed: Connection refuse
d (after 0ms in state CONNECT)
%3|1758680602.436|FAIL|rdkafka#producer-1| [thrd:localhost:9092/bootstrap]: l
ocalhost:9092/bootstrap: Connect to ipv6#[::1]:9092 failed: Connection refuse
d (after 1ms in state CONNECT)
%3|1758680605.455|FAIL|rdkafka#producer-2| [thrd:localhost:9092/bootstrap]: l
ocalhost:9092/bootstrap: Connect to ipv4#127.0.0.1:9092 failed: Connection re
fused (after 0ms in state CONNECT)
%3|1758680606.449|FAIL|rdkafka#producer-1| [thrd:localhost:9092/bootstrap]: l
ocalhost:9092/bootstrap: Connect to ipv4#127.0.0.1:9092 failed: Connection re
fused (after 0ms in state CONNECT)
%3|1758680608.456|FAIL|rdkafka#producer-1| [thrd:localhost:9092/bootstrap]: l
ocalhost:9092/bootstrap: Connect to ipv6#[::1]:9092 failed: Connection refuse
d (after 0ms in state CONNECT)
%3|1758680609.461|FAIL|rdkafka#producer-1| [thrd:localhost:9092/bootstrap]: l
ocalhost:9092/bootstrap: Connect to ipv4#127.0.0.1:9092 failed: Connection re
fused (after 0ms in state CONNECT)
```

Kafka Producer looking for connection

```
{
  "detail": "Database connection failed: 1045 (28000): Access denied for user 'root'@'localhost' (using password: YES)"
}
```

API request failed due to MySQL authentication issue

Action Plan

- Finalize decision regarding Kafka
  - ~~Setup on EC2 Server or locally (?)~~
  - Research Tailscale as a possible choice
- Setup HSML-API and HSML-Web-Form using Docker
  - Previously setup locally
- MySQL Database
  - Fix the database connection
  - Test endpoints
- Setup Kafka and HSML API on the RealVNC Computer

Next Step

1. Setup servers on the RealVNC computer